Blessed G/Michael Catholic Secondary School

1st Semester Mathematics final examination for Grade 12th

Jan., 2018 E.C Time Allowed 2:20 min'

Name Hanna Ayalew Section _____ No _____

I. Choose the best answer among the given alternatives.

1. Which one of the following is an arithmetic sequence

A. $a_n = \left(\frac{1}{2}\right)^n$ B. $a_n = 3n + 1$ C. $a_n = (-1)^n$ D. $a_n = 2^n$

2. The total number of integers between 10 and 200 divisible by 3 is _____

A. 64 B. 46 C. 63 D. 52

3. If the nth term of an arithmetic sequence is A and its $(2n)^{th}$ term is B, Which one is the $(3n)^{th}$ term of the sequence in interms of A and B _____

A. 2B-A B. 2B+2A C. 2B+A D. 2A-B

4. Which one of the following is the first term of the sequence, if the 7th term of an arithmetic sequence is 23, the nth term is 43 and the $(2n)^{th}$ term is 91.

A. 6 B. 8 C. 4 D. 6

5. $\sum_{n=1}^{n} 3\left(\frac{1}{n(n+1)}\right)$ is equal to _____

A. $\frac{400}{201}$ B. $\frac{600}{201}$ C. $\frac{500}{201}$ D. $\frac{201}{300}$

6. The sum of the first 10th term of the sequence $a_n = \ell n\left(\frac{n+1}{n}\right)$ is _____

A. $\ell n12$ B. $\ell n10$ C. $\ell n11$ D. $\ell n9$

7. Which one of the following is not true

A. $8+11+14+\ldots\ldots+98 = \sum_{n=1}^{11}(3n+5)$

B. $6+11+16+\ldots+161 = \sum_{n=1}^{32}(5n+1)$

C. $\frac{1}{2}+\frac{2}{3}+\frac{3}{4}+\ldots\ldots+\frac{30}{31} = \sum_{n=1}^{30}\left(\frac{n}{n+1}\right)$

D. $7+12+17\ldots\ldots \sum_{n=1}^{\infty} 5n+1$

8. The sum of all two digit number between 20 and 100 which are divisible by 6 is _____

A. 780 B. &60 C. 670 D. 880

9. The partial sum of an arithmetic sequence of an= -2n+1 is _____.

A. 2n+1 B. -n2 C. 2n2 D. $\frac{n^2}{2}$

10. When n- arithmetic means are inserted between 12 and 18. The sum of the resulting term is 60, which one of the following is the value of n.

A. 4 B. 8 C. 10 D. 400

11. A man saves birr 400 of his salary in acertain year and in each subsequent year he will save birr 200 more than the previous year. How much will be save during the 8th year?

A. 1800 B. 1400 C. 1200 D. 400

12. The first term of the geometric sequence is 27 and the common ratio of the sequence is $\frac{1}{3}$, the 4th term of the sequence is _____

A. $\frac{1}{9}$ B. $\frac{1}{3}$ C. 1 D. 9

13. If the fourth term of ageometric sequence is 3 and the six term of the sequence is 9, the first term of the sequence is _____

A. $\sqrt{3}$ B. $\frac{\sqrt{3}}{3}$ C. $\frac{1}{9}$ D. $\frac{1}{3}$

14. Which one of the following is the value of x, if x-1, x-3 and x-7 are consecutive terms of ageometric sequence is _____

A. 1 B. 2 C. -2 D. -1

15. if the first term is 5 and the common ratio is 2, the sum of the first 8 terms is _____

A. 1275 B. 12,265 C. 1240 D. 440

16. The geometric mean of 4 and 100 is _____

A. 40 B. 10 C. 20 D. 30

17. How many terms are in the geometric sequence 2,4,8.... 1024?

A. 15 B. 10 C. 20 D. 8

18. If the 3rd term of ageometric sequence is 12 and the 6th term is 96, the common rato of the sequence is ____

A. 4 B. 2 C. 6 D. 10

19. The sum of the first n- natural number is 210, the value of n is _____

A. 40 B. 10 C. 20 D. 30

20. The sum of the first three terms of ageometric sequence is 8 and the sum from 4th to 6th term is 216 common ratio of the sequence is _____

A. 3 B. 4 C. 2 D. 1

21. Which one of the following is convergent sum

A. $\sum_{n=1}^{\infty} 2^n$ B. $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ C. $\sum_{n=1}^{\infty} (-1)^n$ D. $\sum_{n=1}^{\infty} (-2)^n$

22. Which one of the following is convergent sum

A. $\sum_{n=1}^{\infty} 2^n$ B. $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ C. $\sum_{n=1}^{\infty} (-1)^n$ D. $\sum_{n=1}^{\infty} (-2)^n$

23. Which one of the following is divergent sequence

A. $\left(\frac{2}{3}\right)^n$ B. $(-2)^n$ C. $\left(\frac{1}{3}\right)^n$ D. $(\sqrt{2})^n$

24. $\sum_{n=1}^{\infty} (-1)^n \left(\frac{2}{3}\right)^n (-1)$ is equal to _____

A. $\frac{1}{2}$ B. $\frac{2}{5}$ C. $\frac{2}{3}$ D. $\frac{3}{5}$

25. Which one of the following is not true

A. $0.\overline{3} = \frac{1}{3}$ B. $0.\overline{47} = \frac{47}{99}$ C. $\left(\frac{1}{2}\right)^n$ = divergent D. $(-1)^n$ = vibrate

26. The rate change of (-1,3) and (5,9) is _____

A. 4 B. 1 C. 2 D. 3

27. The average rate of change of f(x) = x3+2x+1 on [0,2] is _____

A. 6 B. 4 C. 3 D. 2

28. The gradient of the tangent line to the graph of $f(x)=\sqrt{x-2}$ at $x=6$

A. $\frac{1}{2}$ B. $\frac{1}{3}$ C. $\frac{2}{5}$ D. $\frac{1}{4}$

29. Which one of the following is true

A. $f(x)=2x^{-5}, f'(0)=2$ ✗

B. $f(x)=3x^4, f'(1)=12$ ✓

C. $f(x)=3\sqrt{x}, f'(1)=\frac{2}{3}$ ✗

D. $f(x)=x\sqrt{x}, f'(0)=1$ ✗

30. Let $f(x)=\left(\sqrt{x^3+2x^2+7x}\right)^3$ *then* $\frac{dy}{dx}$ at $x=1$ is _____

A. $21\sqrt{10}$ B. $2\sqrt{3}$ C. *undefined* D. 0

31. Let $y=\sqrt{2+x^4}$, *then the value of* $f'(1)$ *is* _____

A. $2\sqrt{3}$ B. $\frac{2\sqrt{3}}{3}$ C. $\frac{2}{3}$ D. 1

32. What is the equation of straight line having slope -4 that is tangenet to the graph $y=3x^2+2x+1$ s _____

A. $y=4x$ B. $y=4x+1$ C. $y=-4x-2$ D. $y=-4x+6$

33. Let $f(x)=-3x^4+4x^3+12x^2+1$, then the local minimum point of the function is _____

A. (-2,-33) B. (-1,4) C. (0,1) D. (2,33)

34. Which one of the following is the value of a, for which the function ,will have extreme value of $f(x)=\frac{x+3}{x^2+a}$, $atx=2$

A. 4 B. 14 C. 12 D. 16

35. A manfacture wants to design an open box having a square base and a surface area of 108m2. What dimensions will produce a box with maximum volume?

A. 6 and 3 B. 4 and 2 C. 1 and 4 D. 6 and 2

36. Air is being pumped in to spherical ballon at rate of 2cm /min, the rate of change of the radius when the radius is 4cm is _____

A. $\frac{1}{4\pi}$cm/min B. $\frac{1}{32\pi}$cm/min C. $4\pi cm/\min$ D. $1cm/min$

II. Short Answer

37. A 12m ladder is leaning against a well of house if the top of the ladder slices down the wall at a rate of 4cm/sec. how fast will the base be moving away from the wall when 6m from the wall is _____

38. If the demand equation for acertain product is $p=8-\frac{1}{4}x$ *birr* . where x represents the amount of product what is the level of production which results in maximum revenue _____

39. find the area of the region between the graph of $f(x)=1-x^2$ and $g(x)=2-x$ for $x=-1$ and 2 is _____

40. $\int 100dx=$ _____

41. $\int \frac{dy}{dx}(x^3+4x+1)dx=$ _____

42. find the area under the graph of $y=x^3-x$ and x- axis

43. $\int_{-1}^{1} x(x^2+1)^3\,dx=$ _____

44. Find the area s which is enclosed between the graph of $f(x)=(x-1)(x+2)(x)$ and x- axis?

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Name Hanna Aylew Section 12th C No 24

1. B	10. A	19. C	28. D
2. C	11. A	20. A	29. B
3. X	12. C	21. B	30. A
4. C	13. AX	22. B	31. B
5. B	14. D	23. D	32. C
6. C	15. A	24. B	33. A
7. A	16. C	25. C	34. D
8. A	17. B	26. B	35. A
9. B	18. B	27. A	36. B

Short Answer

37, 32 cm²

38, 64.

39, $27/6 = 9/2$.

40, $100x + C$

41, $\frac{x^4}{4} + 2x^2 - 7x + C$

42, $1/2$.

43, $\frac{x^8}{8} + \frac{x^6}{2} + \frac{3x^4}{4} + \frac{x^2}{2} \Big|_{-1}^{+1} = 0$

44. $17/12$

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